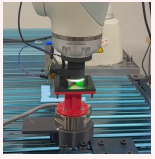
(a) Register real contacts



contact state q^* is read from the robot pose

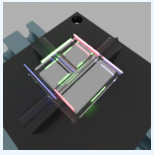
kinematic, mounting and placement errors bias q^*; redo for each session or sensor

project mesh at q^*



cost: one registered pose per label

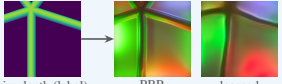
(b) Simulate contacts



mesh gives free geometry; appearance must be rendered

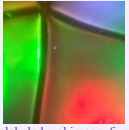
camera, LEDs, gel and contact model calibrated per sensor, or a renderer fitted on real pairs

render



cost: calibration or real pairs; appearance gap remains

(c) AlignTac: align the mesh to the real image



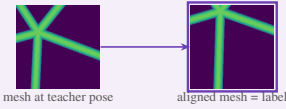
unlabeled real image, fixed

learned from the K registered contacts by perturbing their q^*



aligner $\mathcal{A}$ moves the mesh state q

align



cost: K registered contacts; the rest unlabeled